

Programming in Modern C++: Assignment Week 1

Total Marks : 25

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Question 1

Consider the following code segment.

[MCQ, Marks 2]

```
#include<iostream>
#include<string>
using namespace std;

int main(){
    string greet = "Welcome Back";
    _____; // LINE-1
    cout << greet;
    return 0;
}
```

Choose the correct option to fill in the blank at LINE-1 so that the output is `Welcome`.

- a) ~~greet.resize(7)~~
- b) ~~greet.shrink_to_fit()~~
- c) ~~greet.copy("Welcome", 7, 0)~~
- d) ~~strncpy(greet, "Welcome", 7)~~

Answer: a)

Explanation: `resize(7)` shortens the string to "Welcome". Other functions are not suitable for `std::string` modification or give errors.

Question 2

Consider the following code segment.

[MCQ, Marks 2]

```
#include<iostream>
#include<algorithm>
using namespace std;

bool descending(int i, int j){
    return (i > j);
}

int main(){
    int arr[] = {8, 3, 5, 9, 6, 7};
    sort(_____); // LINE-1
    for(int i=0; i<6; i++)
        cout << arr[i] << " ";
    return 0;
}
```

Choose the correct option to fill in the blank at LINE-1 so that the output is 8 5 3 9 6 7.

- a) arr, arr+3, descending
- b) &arr[0], &arr[0]+4, descending
- c) arr+1, arr+4, descending
- d) &arr, &arr+3, descending

Answer: a)

Explanation: Sorting only a portion of the array, first 3 or positions 1–3.

Question 3

Consider the following code and find the correct output.

[MCQ, Marks 2]

```
#include<iostream>
#include<vector>
#include<algorithm>
using namespace std;

int main(){
    vector<int> nums = {10, 20, 30, 40, 50, 60, 70};
    rotate(nums.begin(), nums.begin()+2, nums.begin()+4);
    for(auto i : nums)
        cout << i << " ";
    return 0;
}
```

- a) 30 10 20 40 50 60 70
- b) 30 40 10 20 50 60 70
- c) 40 10 20 30 50 60 70
- ~~d) 10 30 20 40 50 60 70~~

Answer: b)

Explanation: rotate rearranges elements so that the 3rd becomes the first.

Question 4

Consider the following code and find the correct output or error.

[MCQ, Marks 2]

```
#include<iostream>
#include<stack>
using namespace std;

int main(){
    int arr[5] = {6, 2, 5, 4, 1};
    stack<int> s;
    for(int i=0; i<5; i++){
        while(!s.empty() && s.top() >= arr[i])
            s.pop();
        if(s.empty())
            cout << -1 << " ";
        else
            cout << s.top() << " ";
        s.push(arr[i]);
    }
    return 0;
}
```

- a) -1 6 2 2 1
- b) -1 -1 6 2 2
- c) -1 -1 2 2 -1
- d) Compilation Error

Answer: c)

Explanation: Finds previous smaller element using monotonic stack.

Question 5

Consider the following code segment.

[MSQ, Marks 2]

```
#include<iostream>
using namespace std;

int sumRows(_____) {
    // returns the sum of first row
}

int main(){
    int mat[3][3] = {{1,2,3}, {4,5,6}, {7,8,9}};
    cout << sumRows(mat);
    return 0;
}
```

Choose the correct option to fill in the blank at LINE-1 so that the program does not generate any compilation error.

- a) int mat[] []
- b) int mat[] [3]
- c) int mat[3] []
- d) int mat[3] [3]

Answer: b), d)

Explanation: All but the first dimension must be provided for 2D arrays in function parameters.

Question 6

Consider the following code and choose the correct output.

[MCQ, Marks 2]

```
#include<iostream>
using namespace std;

int main(){
    bool x = false, y = true, z = true;
    cout << (x && y || z);
    return 0;
}
```

- a) 0
- b) 1
- c) false
- d) true

Answer: b)

Explanation: false && true || true evaluates to true = 1

Question 7

Consider the following code segment.

[MSQ, Marks 2]

```
#include <iostream>
#include <string>
using namespace std;

int main() {
    string word1 = "Good ";
    string word2 = "Morning";
    ----- // LINE-1
    cout << message;
    return 0;
}
```

Fill in the blank at LINE-1 so that the program generates output as Good Morning.

- a) `string message = word1 + word2;`
- b) `string message = strcat(word1, word2);`
- c) `string message = word1.insert(word2);`
- d) `string message = word1.append(word2);`

Answer: a), d)

Explanation: Use + or `append()` to concatenate strings.

Question 8

Consider the following code segment.

[MCQ, Marks 2]

```
#include <iostream>
#include <algorithm>

int main() {
    int data[] = {5, 10, 15, 20, 25};
    int key = 20;
    if (_____) // LINE-1
        std::cout << "Found";
    else
        std::cout << "Not Found";
    return 0;
}
```

Fill in the blank at LINE-1 so that the program generates output as Found.

- a) `binary_search(data, data+3, key)`
- b) `std::binary_search(data, data+5, key)`
- c) `std::binary_search(data, data+3, key)`
- d) `binary_search(data, data+5, key)`

Answer: b)

Explanation: Use full range (`data+5`) with qualified namespace.

Question 9

Consider the following code and choose the correct output.

[MCQ, Marks 2]

```
#include <iostream>
#include <algorithm>
using namespace std;

int main() {
    int data[] = {2, 3, 3, 4, 5, 5, 6};
    replace(data, data+7, 3, 5);
    for (int i = 0; i < 4; ++i)
        cout << data[i] << " ";
    return 0;
}
```

- a) 2 4 6 5
- b) 2 4 6 3
- c) 2 5 5 4
- d) 2 5 6 6

Answer: a)

Explanation: After replacement of 3 with 5, the effective sequence becomes 2 5 5 4.

Programming Questions

Question 1

Fill in the blanks at LINE-1 and LINE-2 to implement a function that compares two strings s1 and s2. The function should return: -1 if s1 is shorter, 0 if both strings have equal length, 1 if s1 is longer. *[Marks 3]*

```
#include <iostream>
#include <string>
using namespace std;

int lenCompare(string s1, string s2){
    if (_____) // LINE-1
        return -1;
    else if (_____) // LINE-2
        return 0;
    else
        return 1;
}

int main(){
    string s1, s2;
    cin >> s1 >> s2;
    int res = lenCompare(s1, s2);
    if(res == 0)
        cout << "Equal length";
    else if(res == 1)
        cout << s1 << " is longer";
    else
        cout << s2 << " is longer";
    return 0;
}
```

Public 1

Input: hello worldwide

Output: worldwide is longer

Public 2

Input: sunshine darkness

Output: Equal length

Private

Input: C++ Language

Output: Language is longer

Answer:

- LINE-1: s1.length() < s2.length()
- LINE-2: s1.length() == s2.length()

Question 2

You are given a function to sort characters of a string in descending alphabetical order using a custom comparator. Fill in the blanks to complete the `compareChars` function and make it work with `sort()`. *[Marks 2]*

```
#include <iostream>
#include <algorithm>
#include <string>
using namespace std;

----- compareChars(char a, char b) { // LINE-1
    return (-----);                // LINE-2
}

int main(){
    string str;
    cin >> str;
    sort(str.begin(), str.end(), compareChars);
    cout << str;
    return 0;
}
```

Public 1

Input: delta

Output: tleda

Public 2

Input: apple

Output: pplea

Private

Input: okay

Output: yoka

Answer:

- LINE-1: `bool`
OR
LINE-1: `int`
- LINE-2: `a > b`

Question 3

Complete the functions to: Replace all instances of value x with y in the array. Then reverse the array. *[Marks 2]*

```
#include <iostream>
#include <algorithm>
using namespace std;

void replaceValue(int arr[], int n, int x, int y){
    -----; // LINE-1
}

void reverseArray(int arr[], int n){
    -----; // LINE-2
}

int main(){
    int arr[] = {1, 4, 3, 4, 2, 4};
    int x, y;
    cin >> x >> y;
    int n = sizeof(arr)/sizeof(arr[0]);

    replaceValue(arr, n, x, y);
    for(int i = 0; i < n; i++)
        cout << arr[i] << " ";
    cout << endl;

    reverseArray(arr, n);
    for(int i = 0; i < n; i++)
        cout << arr[i] << " ";
}
```

Public 1

Input: 4 9

Output:

1 9 3 9 2 9

9 2 9 3 9 1

Public 2

Input: 3 5

Output:

1 4 5 4 2 4

4 2 4 5 4 1

Private

Input: 1 9

Output:

9 4 3 4 2 4

4 2 4 3 4 9

Answer:

- LINE-1: `replace(arr, arr + n, x, y);`
- LINE-2: `reverse(arr, arr + n);`